AKTU NEXTGEN ENGINEERING

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FREE COURSES

SOFTWARE ENGINEERING

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Unit - 5

ONE SHOT

3 PYQ Solutions

⇒ Topics :

- ☆ Software as an Evolutionary Entity,
- ☆ Need for Maintenance
- ☆ Categories of Maintenance :
  - → Preventive
  - → Corrective
  - → Perfective
- ☆ Cost of Maintenance.
- ☆ Software Re-Engineering and Reverse Engineering
- ☆ Software Configuration Management Activities
- ☆ Change Control Process
- ☆ Software version Control
- ☆ Resource Allocation Models
- ☆ Constructive Cost Models
- ☆ Software Risk analysis and Management

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⇒ Software Maintenance : [AKTU - 2021-22]

→ Software maintenance is part of Software Development life cycle.
→ Its main purpose is to modify and update software application after delivery to correct faults and to improve performance.
→ The essential part of software maintenance requires preparation of an accurate plan during the development cycle.
→ The cost of software maintenance is high.

⇒ Need for Software maintenance

- Over a period of time, software's original requirements may change to reflect the customer need.
- Correct faults
- Improve the design
- Implement enhancement
- Interface with other system
- Requirement of user change
- Run the code fast

⇒ Why Software Maintenance required?

- Bug Fixing : The process of finding and fixing errors and problems in the software
- Enhancement :- The process of adding new feature or improving existing features to meet the evolving need of the users

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- Performance Optimization :- The process of improving the speed, efficiency and reliability of the software.
- Porting and Migration :- The process of adapting the software to run on new hardware or software platform.
- Re-Engineering :- The process of improving the design and architecture of the software to make it more maintable and scalable.
- Documentation :- The process of creating, updating, and maintaining the documentation for the software.

⇒ Categories of Software Maintenance with Example. [AKTU - 201-22]

- Corrective maintenance : Corrective maintenance of a software product may be essential either to rectify some bugs observed while the system is in use, or to enhance the performance of the system.
- This type of maintenance is called bug fixing.
  Ex:- A piece of software may crash when being used with another piece of software.
- Adaptive maintenance :- This includes modification and updation when the customers need the product to run on new platforms, on new operating systems, or when they need the product to interface with new hardware and software.

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Ex:- A virus threat / hacker threat means that the software will need to be adapted to protect against this.

- Perfective maintenance: A software product need maintenance to support the new features that the users want or to change different types of functionalities of the system according to the customer's demands.

Ex: The software provider provides upgrades which will improve the performance of the software.

- Preventive maintenance:- This type of maintenance include modifications and updations to prevent future problems with the software.

→ It goals to attend to problems, which are not significant at this moment but may cause serious issues in the future.

Ex: Common examples of preventive maintenance tasks are regular cleaning, lubrication, replacing of parts and equipment repair.

⇒ Cost :-

- The cost of system maintenance represents a large proportion of the buget of most organizations that use software system

- More than 65% of software lifecycle cost is expended in the maintenance activities.

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⇒ Cost of software maintenance can be control by postponing the development opportunity of software maintenance but this will cause following intangible cost:

- Customer dissatisfaction when inquiries for repair or modification cannot be addressed in a timely manner.
- Reduction in overall software quality as a result of changes that introduce hidden errors in maintained software

⇒ Efforts:-

- Efforts expended on maintenance may be divided into productivity activities and evaluation, design and modification, coding
- The following expression provides a model of maintenance efforts

$$M = P + K(C-D)$$

where M = total effort expended
P = productive effort
K = An empirical constant.
C = measure of complexity that can be attributed a lack of good design
D = A measure of degree of familarity with the software.

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⇒ Draw the general model of Software re-engineering and elaborate it. [AKTU - 2022-23]

⇒ Software Re-engineering is a process of software development that is done to improve the maintainability of a software system.

⇒ Re-engineering is the examination and alteration of system to reconstitute it in a new form.

⇒ This process encompasses a combination of sub processes like reverse engineering, forward engineering, reconstructing, etc.

⇒ It affect positively at software cost, quality service to the customer and speed of delivery.

⇒ In software re-engineering we are improving the software to make it efficient and effective.

Objective :-

- To describe a cost-effective option for system evolution
- To describe the activities involved in the software maintenance process
- To distinguish between software and data re-engineering and to explain the problem of data re-engineering.

Why Perform Re-engineering :-

- To improve the software performance and scalability
- To add new features
- To support new platforms
- To improve maintainability

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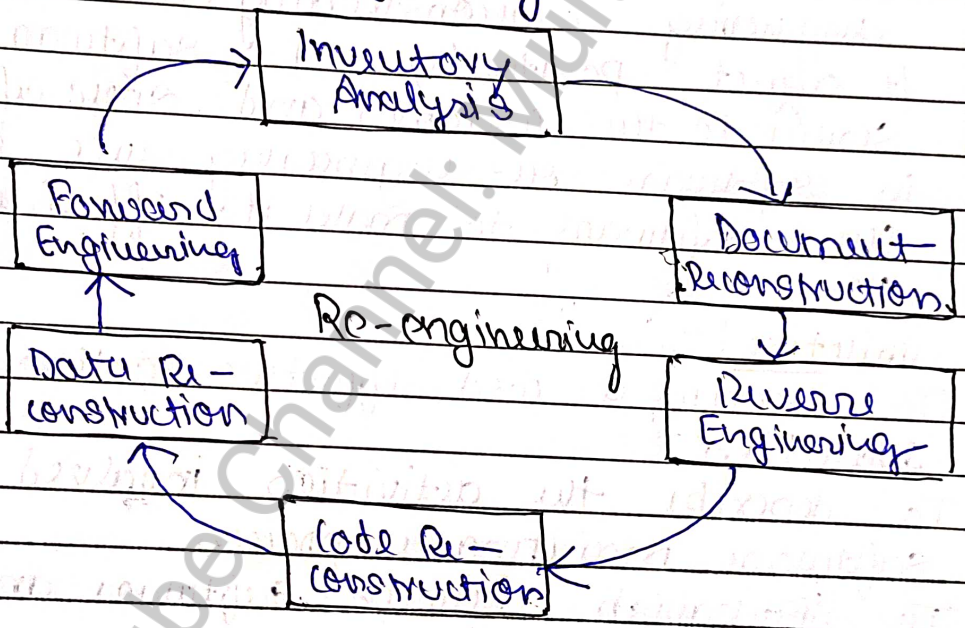
- To meet new regulations and compilance
- Improve saftwar quality.
- Updating technalogy, resalving issues.

⇒ Steps invalued in Re-engineering

1. Inventory Analysis
2. Documut Reconstruction
3. Reverse Engineering
4. Code Reconstruction
5. Data Reconstruction
6. Forward Engineering



⇒ Inventory Analysis:

- Every saftware organization should have an inventory of all applications.
- The inventory can be nothing more than a spreadsheet model containing information that provides a detailed description of every active application

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- Resources can be allocated to candidate application for re-engineering work.

⇒ Document restructuring:

- Documentation of a system either explains how it operates or how to use it.
- Documentation must be updated.
- It may not be necessary to fully re-document an application.
- Rather those portions of the system that are current undergoing change are fully documented
- Over time a collection of useful and relevant documentation will evolve.
- The system is business critical and must be fully re-documented

⇒ Reverse engineering:

- Reverse engineering is a process of design recovery
- Reverse engineering tool extract data, architectural and procedural design information from an existing program

⇒ Code reconstructing:

- To accomplish code restructuring the source code is analyzed using a restructuring tool.
- Violations of structured programming constructs are noted and code is then restructured.
- The resultant restructured code is reviewed and tested to ensure that no anomalies have been introduced.
- Internal code documentation is updated

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⇒ Data restructuring :
- Data restructuring begins with a reverse engineering activity.
- Current data structure is dissected and necessary data models are defined.
- Data objects and attributes are identified, and existing data structure are reviewed for quality.

⇒ Forward engineering :
- Forward engineering also called renovation or reclamation not only recovers design information from existing software, but uses this information to alter or reconstitute the existing system in an effort to improve its overall quality.

Adv
- Reduced Risk
- Reduced cost
- Revelation of Business Rules
- Improved efficiency
- Increased flexibility
- Improved quality

Disadv :-
- High cost
- Risk of failure
- lack of employee involvement.
- Resistance to change
- Difficulty in measuring success

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Reverse - Engineering :-

- Reverse Engineering tools extract data, architectural and procedural design information from an existing program.
- Software Reverse Engineering is the process of recovering the design and the requirements specification of a product from an analysis of its code.
- Reverse engineering is becoming important since several existing software products lack proper documentation, are highly unstructured.
- The aim of reverse engineering is to improve the understandability of the system by helping the maintenance work.

Objective :-

- Reducing costs
- Analysis of security
- Integration and customization
- Recovering lost source code.
- fixing bugs and maintenance.

Steps of software Reverse Engineering

- Collect information: This step focuses on collecting all possible information about the software

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- Examining the information: The information collected in step-1 is studied so as to get familiar with the system
- Extracting the structure: This step concerns identifying program structure in the form of a structure chart where each node correspond to some routine.
- Recording the Functionality:- During this step processing detail of each module of the structure charts are recorded using structured language like decision table, etc.
- Recording data flow:- From the information extracted in step 3 and step 4, a set of data flow diagrams is derived to show the flow of data among the process
- Recording control flow:- The high-level control structure of the software is recorded.
- Review Extracted Design:- The design document extracted is reviewed several times to ensure consistency and correctness
- Generate documentation:- Finally, in this step, the complete documentation including SRS, design, document, history, overview etc is recorded for future use.

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⇒ Re-engineering :-

- It is a process to re-design a product or a particular component.
- It is also called as software refactoring.
- It is like a modification of the existing product to improve its performance, maintainability and adaptability.
- It is a cost effective method as a part of the software is altered instead of the entire product.
- Less coding knowledge is required for re-engineering.
- This practice is generally followed for safety issues or to update a new features into the product existing functionality.
- Re-engineering take less time in comparison to Reverse Engineering.

⇒ Reverse Engineering :-

- It is a process of deconstruction a product to see how it actually works.
- It is also known as backward engineering.
- In this, the product is broken down to see its functionality and is then iterated to bring something new out of it.
- It is an expensive method as the entire product is rebuilt.
- It require prior coding knowledge regarding the software followed by knowledge of system components and their relationships.

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=> This practice is generally followed by companies to evaluate their competitors latest inventions and get something better out of it.

=> This process comparatively takes more time as the product is created from scratch

=> Software Configuration Management [2022-23]

=> It can be defined as a process of defining and implementing a standard configuration, which results into the primary benefits such as easier setup and maintenance, less down-time, better integration with enterprise management, and more efficient and reliable backups and also maximize productivity by minimizing mistakes.

=> SCM is used to track and manage the emerging product and its version

=> SCM ensures that all people involved in the Software process know what is being designed, developed, built, tested and delivered.

=> Through SCM, the design requirements can be traced to the final software product

Objective :-

- Remote system administration
- Reduced user down-time
- Reliable data backup

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- Early workstation setup
- Multi-user support

⇒ Task in SCM process

- Configuration Identification :-
→ It is a method of determining the scope of the software system.
→ With the help of this, step you can manage or control something even if you don't know what it is.

- Baseline :-
→ A baseline is a formally accepted version of a software configuration item.
→ It is designated and fixed at a specific time while conducting the SCM process.
→ It can only be changed through formal change control procedures.

- Change Control :-
→ Change control is a procedural method which ensures quality and consistency when changes are made in the configuration object.
→ In this step, the change request is submitted to SCM.

- Configuration Status Accounting :
→ Configuration status accounting tracks each release during the SCM process.
→ This stage involves tracking what each version

has and the changes that lead to this version.

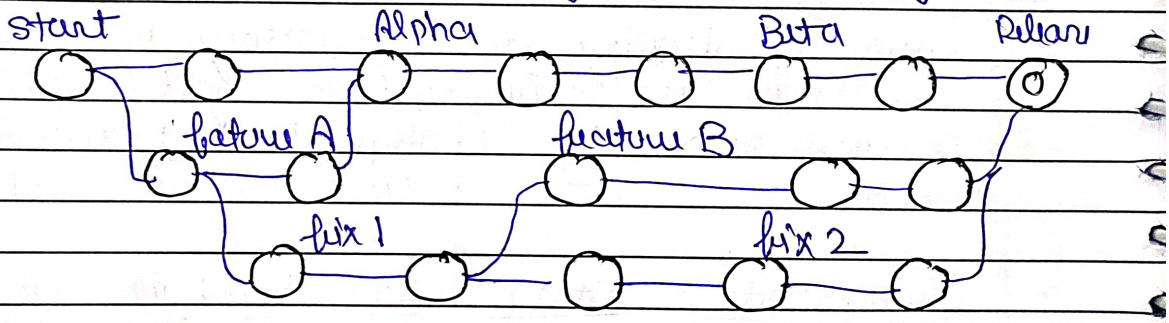
- Configuration Audits and Reviews:

⇒ Software Configuration audits verify that all the software product satisfies the baseline needs.

⇒ It ensures that what is built is what is delivered

⇒ Software version control

- Version control is a way to keep track of changes in computer programs.
- It helps work people together on a project without causing problem.
- They can see what changes were made, when, and by whom.
- It also lets them go back to an older version if something goes wrong.



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⇒ Software version control Activities

- Identifying New Versions:
→ Each time changes are made and saved, the version control system creates a new, unique version of the software.
→ This helps developers track the software's evolution and easily manage multiple versions

- Numbering scheme:
→ Version control systems assign a specific numbering or naming scheme to each version (e.g 1.0, 1.1, 1.2, or 2.0).
→ This helps developer and users identify the software's progression and understand the scope of changes between versions.

- Visibility:- Version control makes the entire history of the software project visible to all team members.
→ This enhances collaboration and helps developers understand the context and impact of past changes on the current codebase.

- Tracking:-
→ Version control system keep detailed records of who made each change, when it was made, and the specific modification.
→ This level of tracking helps maintain accountability and facilitates communication among team member.

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⇒ CASE Tools { AKTU - 2019-20 }

- It stands for Computer Aided Software Engineering.
- It means, development and maintenance of software project with help of various automated software tools.
- CASE tools are set of software application programs, which are used to automate SDLC activities.
- CASE tools are used by software project managers, analyst and engineers to develop software system
- There are a number of CASE tools available to simplify various stages of software Development Life cycle such as Analysis tools, Design tools, Project management tools, DBM tools, Documentation tool etc
- CASE tools helps the project manager, the software developer and other key personel to improve their productivity in the development team

⇒ Architecture of CASE environment

⇒ The important component of a modern CASE environment are :-

1] User interface
2] Tool set
3] Object Management system (OMS)
4] A repository

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⇒ User Interface :- The user interface provided a consistent framework for accessing the different tool thus making it easier for the users to interact with the different tools and hence reducing the learning period.

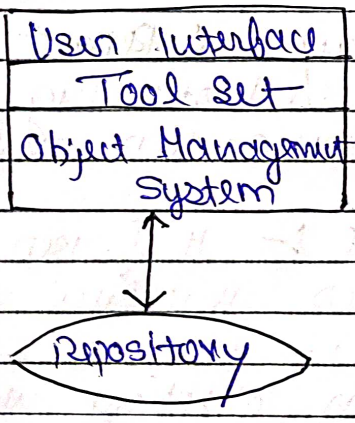
⇒ Object management system :- Different CASE tools represent the software product as a set of entities such as specification, design, test data, project plan etc.

⇒ The object management system maps these logical entities into the underlying storage management system (repository).

⇒ Repository :- The complete information about meta-models, methods, project artifacts, baseline process models etc. is available in the repository.

⇒ This information can be shared by different team members efficiently thereby improving the project management as a whole.

⇒ While implementing, the repository can be either centralized or distributed.



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Benefits :-
1. Improved productivity
2. Better documentation
3. Reduced lifetime maintenance
4. Improved accuracy
5. Intangible benefits

limitation of CASE tools -
1. Cost
2. learning curve
3. Tool mix : It is important to make an appropriate selection of tool mix to get cost advantage

=> Types of CASE tools :-

1. Diagramming Tools :-
-> It helps in diagrammatic and graphical representation of the data and system process.
-> It represents system elements, control flow and data flow among different software component and system structures in a pictorial form

2. Computer Display and Report Generators :-
-> They help in understanding the data requirement and the relationship involved.

3. Analysis tool :- It focuses on inconsistent, incorrect specifications involved in the diagram and data flow.
-> It helps in collecting requirements, automatically

check for any irregularity, imprecision in the diagrams, data redundancies or erroneous omission.

4. Central Repository :- It provides a single point of storage for data diagrams, reports and documents related to project management.

5. Tools for Requirement Management :- It makes gathering, evaluating and managing software needs easier.

6. Tools for Analysis and Design :- It offers instruments for modelling system architecture and behaviour, which helps throughout the analysis and design stages of software development.

7. Tools for Database Management :- It facilitates database construction, design and administration.

8. Tools for Documentation :- It makes the process of writing, organizing, and maintaining project document easier.

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⇒ PERT

- It stands for Program Evaluation Review Technique.
- PERT is a project management planning tool used to calculate the amount of time it will take to realistically finish a project.
- PERT charts are used to plan tasks within a project making it easier to schedule and coordinate team members.
- PERT can be both a cost and a time management system.
- It doesn't not deal very well with task overlap.
- PERT is organized by events and activities or tasks
- Each chart starts with an initiation node from which the first task or tasks originates.
- If multiple tasks begin at the same time, they are all started from the node or branch or fork out from the starting point
- Each task is represented by line which states its name or other identifier or its duration.
- The other end of the task line is terminated by another node, which identifies the start of another task.

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⇒ Benefits :-

- The pert network is continuously useful to project managers prior to and during a project.
- The PERT network graphical representation of the project takes help to show the task interrelationships.
- It allows scheduling and simulation of alternative schedules

limitation :-

→ The PERT network does not deal very well with task overlap.

→ PERT does not help in deciding which activities are necessary or how long each will take.

⇒ Critical Path Method :-

⇒ Critical Path method (CPM) is a method used in project planning, generally for project scheduling for the on-time completion of the project.

⇒ It helps in the determination of the earliest time by which the whole project can be completed.

⇒ There are two main concepts in this method name critical task and critical path.

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Benefits :
- Show the project schedule visually.
- Highlight important tasks with CPM.
- Use CPM to find and handle risk.
- CPM helps the project team communicate better

limitation :—
- The relation of tasks to time is not immediate
- CPM charts are more difficult to understand

⇒ Constructive Cost Model [AKTU 19-20, 21-22, 22-23]

⇒ It was developed by a scientist Barry Boehm in 1981

⇒ The COCOMO (Constructive cost Model) is one of most popularly used software cost estimation models.

⇒ This model depends on the size means number of lines of code for software product development

⇒ It estimates Effort required for project, Total project cost & Scheduled time of project

⇒ In COCOMO, projects are categorized into three types :

1. Organic type :
- → Project is small and simple (2-50 KLOC)
- → Few Requirement of Projects
- → Project team is small with prior experience

→ The problem is well understood and has been solved in the past.

Example:- Simple Inventory Management system and Data Processing system

2. Semidetached type:-
→ Medium size and has mixed rigid requirement (50-300 KLOC)
→ Project has complexity not too high & low.
→ Both experienced and inexperienced member in Project team.
→ Project are few known and few unknown modules

Example:- DBMS, Difficult inventory management system

3. Embedded type:-
→ Large project with fixed requirement of resources (More than 300 KLOC)
→ Larger team size with little previous experience
→ Highest level of complexity, creativity and experienced requirement

Example:- ATM, Air Traffic control, Banking Software

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⇒ Types of COCOMO Model.
- Basic COCOMO Model
- Intermediate COCOMO Model
- Complete / Detailed COCOMO Model.

⇒ Type 1: Basic COCOMO Model.
- It is type of static model to estimates software development effort quickly and roughly.
- It mainly deals with number of lines of code in project.

Formula:-

Effort (E) = $a*(KLOC)^b$ MM

Scheduled Time (D) = $c*(E)^d$ Months(M)

Where
- E = total effort required for the project in man-months (MM).
- D = Total time required for project development in Months (M)
- KLOC = The size of code for the project in kilo lines of code.
- a,b,c,d = The constant parameter for a software project

PROJECT TYPE	a	b	c	d
Organic	2.4	1.05	2.5	0.38
Semi detached	3	1.12	2.5	0.35
Embedded	3.6	1.2	2.5	0.32

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=> Type 2 : Intermediate COCOMO Model

- Extension of Basic COCOMO model which enhance more accuracy to cost estimation model result
- It include cost drivers (Product, Hardware, Resource & Project Parameter) of Project

Formula :-

$$\text{Effort (E)} = a * (\text{KLOC})^b * \text{EAF MM}$$
$$\text{Scheduled Time (D)} = c * (E)^d \text{ Months (M)}$$

where

- E = total effort required for the project in Man-Months (MM)
- D = Total time required for project development in Months (M).
- KLOC = The size of the code for the project in Kilo lines of code.
- a, b, c, d = The constant parameters for the software project.

PROJECT TYPE	a	b	c	d
Organic	2.4	1.05	2.5	0.38
Semi-Structured	3	1.12	2.5	0.35
Embedded	3.6	1.2	2.5	0.32

- EAF : Effort Adjustment factor, which is calculated by multiplying the parameter values of different cost driven parameter. For ideal the value is 1.

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=> Type 3 : Detailed / Complete COCOMO Model

- The model incorporates all qualities of both Basic COCOMO and Intermediate COCOMO strategies on each software engineering process.
- The whole software is divided into different modules and then apply COCOMO in different modules to estimate effort and then sum the effort

-> The six phases of detailed COCOMO are :-

1. Planning and requirement
2. System design
3. Detailed design
4. Module code and test
5. Integration and test
6. Cost Constructive model.

Adv :-

- Provide a systematic way to estimate the cost and effort of a software project
- Estimate cost and effort of software project at different stages of the development process.
- Provide ideas about historical projects
- Easy to implement with various factors

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Disadv :-

- It mostly depend on time factors.
- It is based on assumptions and averages.
- It limits the accuracy of the software cost.
- It ignores requirement, customer skill and hardware issues.

⇒ Risk Analysis and Management :-

- "Tomorrow problem are today's risk.
- Hence a clear definition of a "risk" is a problem that could care some loss or threaten the progress of the project, but which has not happened yet.
- There potential issues might harm cost, schedule or technical success of the project and the quality of our software device, or project team morale.
- Risk Management is the system of identifying addressing and eliminating these problems before they can damage the project.

There square measures 3 main classes of risk :-

1. Project risks
2. Technical risks
3. Business risks

⇒ Project risks : Project risks concern differ forms of budgetary, schedule, personnel, resource and customer-related problems.

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2. Technical risks: Technical risks concern potential method, implementation, interfacing, testing and maintenance issue.

3. Business risks: This type of risks contain risks of building an excellent product that no one need, losing budgetary or personnel commitments etc.

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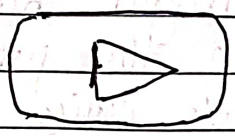

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